

POMDP Formulation Approximations

POMDP Computational Complexity

POMDP Computational Complexity

Sad facts 🤔

POMDP Computational Complexity

Sad facts 🥲

- Infinite horizon POMDPs are *undecidable*

POMDP Computational Complexity

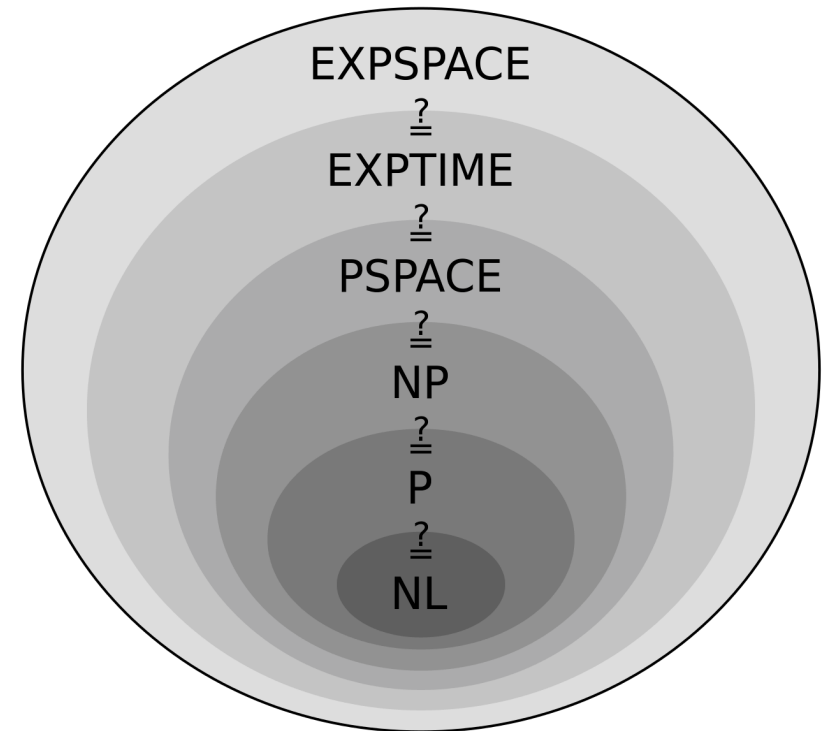
Sad facts 🥹

- Infinite horizon POMDPs are *undecidable*
- Finite horizon POMDPs are *PSPACE Complete*
 - Among the hardest problems that can be solved using a polynomial amount of space

POMDP Computational Complexity

Sad facts 🥹

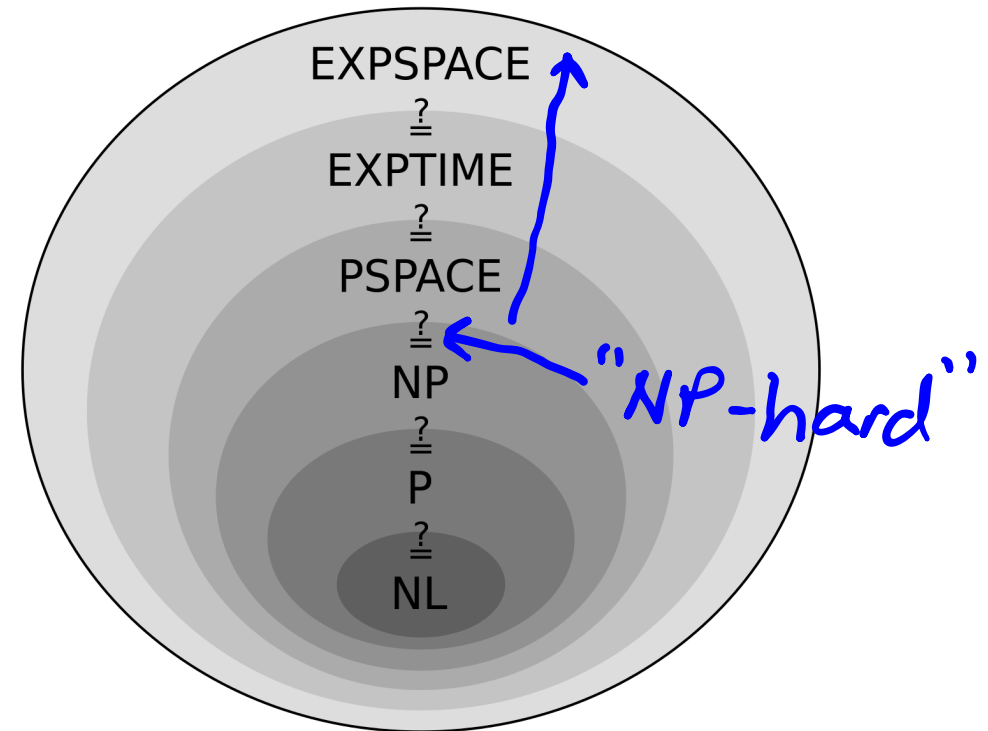
- Infinite horizon POMDPs are *undecidable*
- Finite horizon POMDPs are *PSPACE Complete*
 - Among the hardest problems that can be solved using a polynomial amount of space



POMDP Computational Complexity

Sad facts 🥹

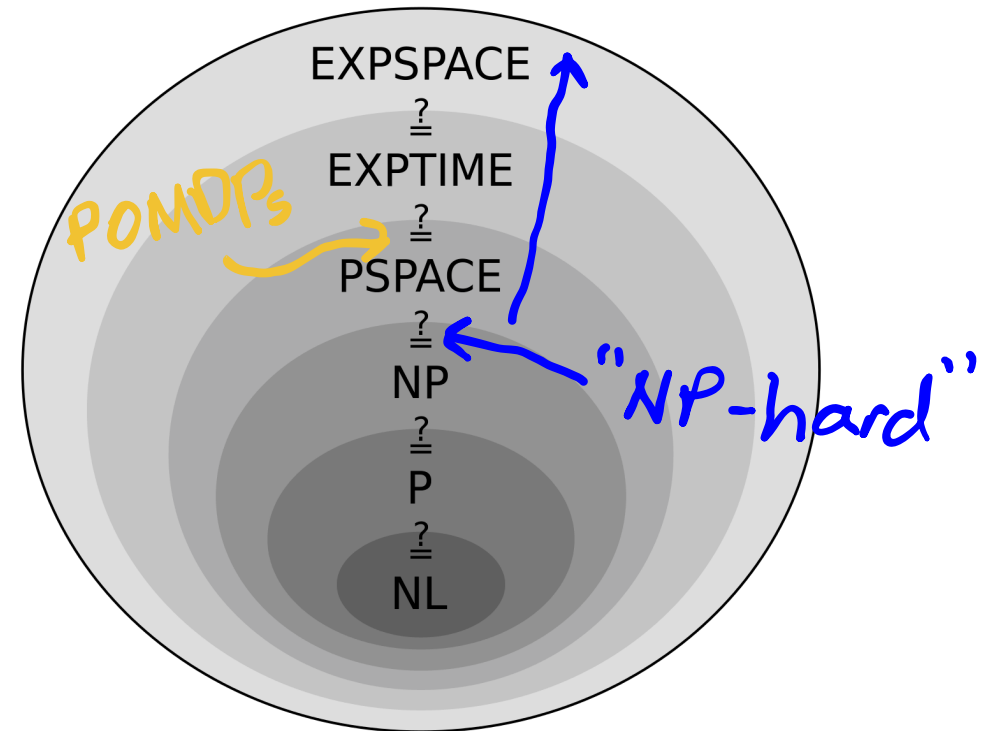
- Infinite horizon POMDPs are *undecidable*
- Finite horizon POMDPs are *PSPACE Complete*
 - Among the hardest problems that can be solved using a polynomial amount of space



POMDP Computational Complexity

Sad facts 🥹

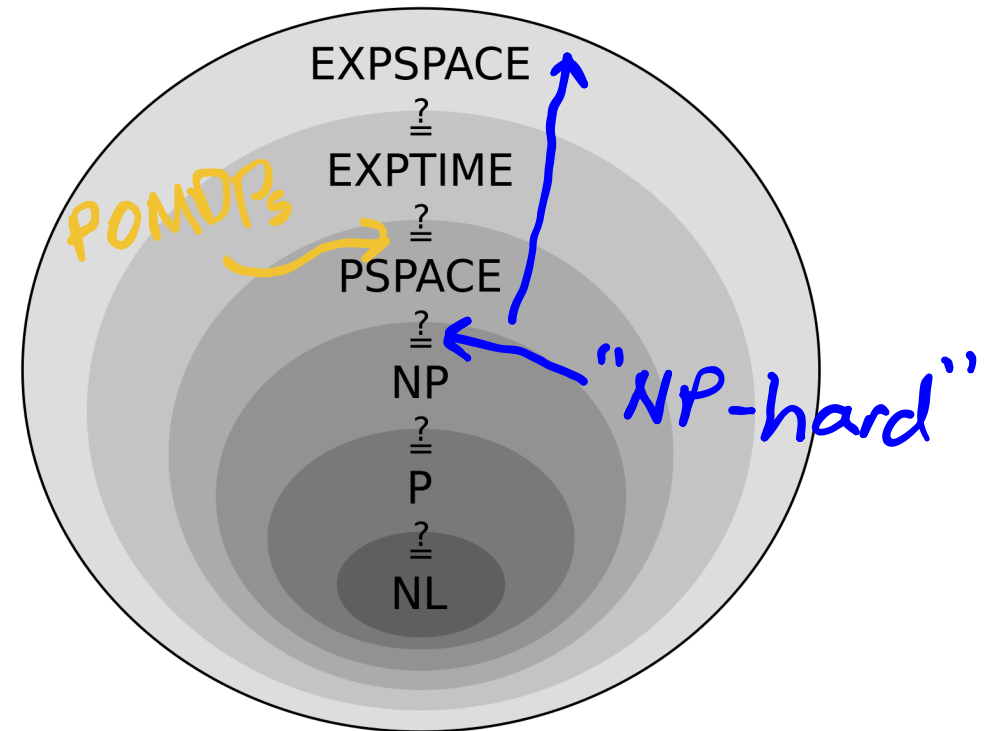
- Infinite horizon POMDPs are *undecidable*
- Finite horizon POMDPs are *PSPACE Complete*
 - Among the hardest problems that can be solved using a polynomial amount of space



POMDP Computational Complexity

Sad facts 🥹

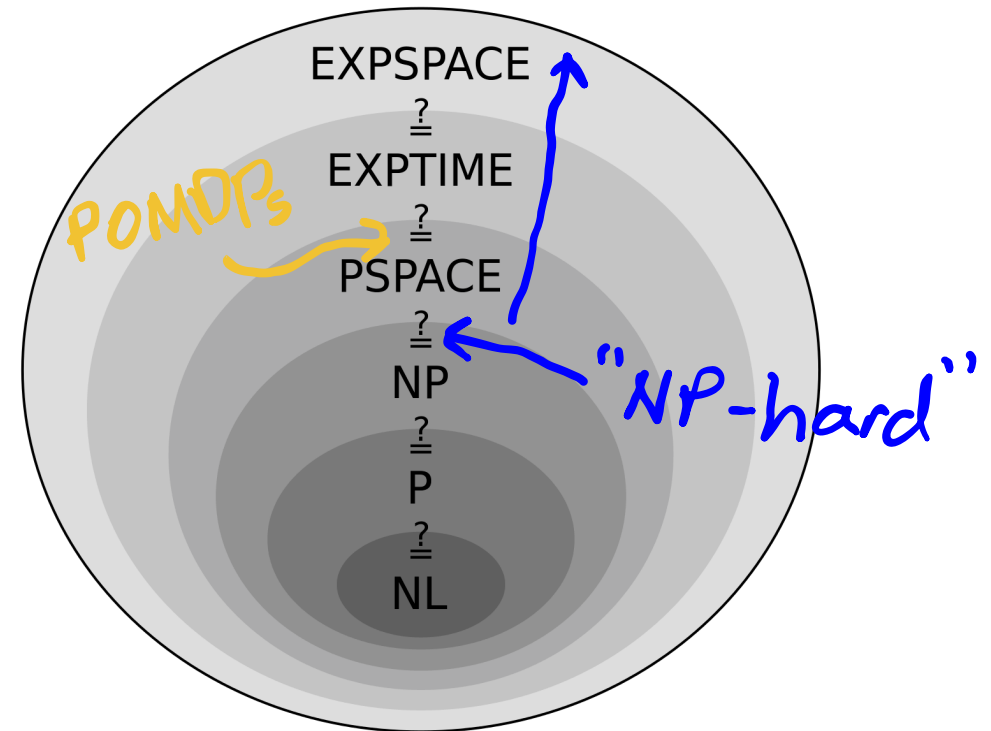
- Infinite horizon POMDPs are *undecidable*
- Finite horizon POMDPs are *PSPACE Complete*
 - Among the hardest problems that can be solved using a polynomial amount of space
 - Any algorithm that can solve a general POMDP will have exponential complexity



POMDP Computational Complexity

Sad facts 🥹

- Infinite horizon POMDPs are *undecidable*
- Finite horizon POMDPs are *PSPACE Complete*
 - Among the hardest problems that can be solved using a polynomial amount of space
 - Any algorithm that can solve a general POMDP will have exponential complexity (we think)



Approximate POMDP Solutions

Approximate POMDP Solutions

Numerical Approximations

(approximately solve original problem)

Approximate POMDP Solutions

Numerical Approximations

(approximately solve original problem)



Offline

Approximate POMDP Solutions

Numerical Approximations

(approximately solve original problem)



Offline

SARSOP

Approximate POMDP Solutions

Numerical Approximations

(approximately solve original problem)



Offline

SARSOP



Online

Approximate POMDP Solutions

Numerical Approximations

(approximately solve original problem)



Offline

SARSOP



Online

(Next time)

Approximate POMDP Solutions

Numerical Approximations

(approximately solve original problem)



Offline

SARSOP



Online

(Next time)

Formulation Approximations

(solve a slightly different problem)

Approximate POMDP Solutions

Numerical Approximations

(approximately solve original problem)



Offline

SARSOP



Online

(Next time)

Formulation Approximations

(solve a slightly different problem)

Today!

POMDP Objective

POMDP Objective

$$\pi^* = \operatorname{argmax}_{\pi: B \rightarrow A} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t)) \right]$$

POMDP Objective

$$\pi^* = \operatorname{argmax}_{\pi: B \rightarrow A} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t)) \right]$$

$$b' = \tau(b, a, o)$$

Certainty Equivalent

POMDP Objective

$$\pi^* = \operatorname{argmax}_{\pi: B \rightarrow A} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t)) \right]$$

$$b' = \tau(b, a, o)$$

Certainty Equivalent

POMDP Objective

$$\pi^* = \operatorname{argmax}_{\pi: B \rightarrow A} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t)) \right]$$

$$b' = \tau(b, a, o)$$

$$b' = \tau(b, a, o)$$

Certainty Equivalent

POMDP Objective

$$\pi^* = \operatorname{argmax}_{\pi: B \rightarrow A} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t)) \right]$$

$$b' = \tau(b, a, o)$$

$$\pi_{\text{CE}}(b) = \pi_s(\hat{s}(b))$$

$$b' = \tau(b, a, o)$$

Certainty Equivalent

Optimal for Linear-Quadratic-Gaussian (LQG)
(Analogous to LQR MDP)

Certainty Equivalent

Optimal for Linear-Quadratic-Gaussian (LQG)
(Analogous to LQR MDP)

LQG POMDP

$$\mathcal{S} = \mathbb{R}^n, \mathcal{A} = \mathbb{R}^m, \mathcal{O} = \mathbb{R}^p$$

$$T(\mathbf{s}' \mid \mathbf{s}, \mathbf{a}) = \mathcal{N}(\mathbf{s}' \mid \mathbf{T}_s \mathbf{s} + \mathbf{T}_a \mathbf{a}, \Sigma_s)$$

$$O(\mathbf{o} \mid \mathbf{s}') = \mathcal{N}(\mathbf{o} \mid \mathbf{O}_s \mathbf{s}', \Sigma_o)$$

$$R(s, a) = -s^\top R_s s - a^\top R_a a$$

Certainty Equivalent

Optimal for Linear-Quadratic-Gaussian (LQG)
(Analogous to LQR MDP)

LQG POMDP

$$b(\mathbf{s}) = \mathcal{N}(\mathbf{s} \mid \boldsymbol{\mu}_b, \boldsymbol{\Sigma}_b)$$

$$\mathcal{S} = \mathbb{R}^n, \mathcal{A} = \mathbb{R}^m, \mathcal{O} = \mathbb{R}^p$$

$$T(\mathbf{s}' \mid \mathbf{s}, \mathbf{a}) = \mathcal{N}(\mathbf{s}' \mid \mathbf{T}_s \mathbf{s} + \mathbf{T}_a \mathbf{a}, \boldsymbol{\Sigma}_s)$$

$$O(\mathbf{o} \mid \mathbf{s}') = \mathcal{N}(\mathbf{o} \mid \mathbf{O}_s \mathbf{s}', \boldsymbol{\Sigma}_o)$$

$$R(s, a) = -s^\top R_s s - a^\top R_a a$$

Certainty Equivalent

Optimal for Linear-Quadratic-Gaussian (LQG)
(Analogous to LQR MDP)

LQG POMDP

$$\mathcal{S} = \mathbb{R}^n, \mathcal{A} = \mathbb{R}^m, \mathcal{O} = \mathbb{R}^p$$

$$T(\mathbf{s}' \mid \mathbf{s}, \mathbf{a}) = \mathcal{N}(\mathbf{s}' \mid \mathbf{T}_s \mathbf{s} + \mathbf{T}_a \mathbf{a}, \Sigma_s)$$

$$O(\mathbf{o} \mid \mathbf{s}') = \mathcal{N}(\mathbf{o} \mid \mathbf{O}_s \mathbf{s}', \Sigma_o)$$

$$R(s, a) = -s^\top R_s s - a^\top R_a a$$

$$b(\mathbf{s}) = \mathcal{N}(\mathbf{s} \mid \boldsymbol{\mu}_b, \Sigma_b)$$

$$\boldsymbol{\mu}_p \leftarrow \mathbf{T}_s \boldsymbol{\mu}_b + \mathbf{T}_a \mathbf{a}$$

$$\Sigma_p \leftarrow \mathbf{T}_s \Sigma_b \mathbf{T}_s^\top + \Sigma_s$$

Certainty Equivalent

Optimal for Linear-Quadratic-Gaussian (LQG)
(Analogous to LQR MDP)

LQG POMDP

$$S = \mathbb{R}^n, A = \mathbb{R}^m, O = \mathbb{R}^p$$

$$T(\mathbf{s}' \mid \mathbf{s}, \mathbf{a}) = \mathcal{N}(\mathbf{s}' \mid \mathbf{T}_s \mathbf{s} + \mathbf{T}_a \mathbf{a}, \Sigma_s)$$

$$O(\mathbf{o} \mid \mathbf{s}') = \mathcal{N}(\mathbf{o} \mid \mathbf{O}_s \mathbf{s}', \Sigma_o)$$

$$R(s, a) = -s^\top R_s s - a^\top R_a a$$

$$b(\mathbf{s}) = \mathcal{N}(\mathbf{s} \mid \mu_b, \Sigma_b)$$

$$\mu_p \leftarrow \mathbf{T}_s \mu_b + \mathbf{T}_a \mathbf{a}$$

$$\Sigma_p \leftarrow \mathbf{T}_s \Sigma_b \mathbf{T}_s^\top + \Sigma_s$$

$$\mathbf{K} \leftarrow \Sigma_p \mathbf{O}_s^\top \left(\mathbf{O}_s \Sigma_p \mathbf{O}_s^\top + \Sigma_o \right)^{-1}$$

$$\mu_b \leftarrow \mu_p + \mathbf{K}(\mathbf{o} - \mathbf{O}_s \mu_p)$$

$$\Sigma_b \leftarrow (\mathbf{I} - \mathbf{K} \mathbf{O}_s) \Sigma_p$$

Certainty Equivalent

Optimal for Linear-Quadratic-Gaussian (LQG)
(Analogous to LQR MDP)

LQG POMDP

$$S = \mathbb{R}^n, A = \mathbb{R}^m, O = \mathbb{R}^p$$

$$T(\mathbf{s}' \mid \mathbf{s}, \mathbf{a}) = \mathcal{N}(\mathbf{s}' \mid \mathbf{T}_s \mathbf{s} + \mathbf{T}_a \mathbf{a}, \Sigma_s)$$

$$O(\mathbf{o} \mid \mathbf{s}') = \mathcal{N}(\mathbf{o} \mid \mathbf{O}_s \mathbf{s}', \Sigma_o)$$

$$R(s, a) = -s^\top R_s s - a^\top R_a a$$

$$b(\mathbf{s}) = \mathcal{N}(\mathbf{s} \mid \mu_b, \Sigma_b)$$

$$\mu_p \leftarrow \mathbf{T}_s \mu_b + \mathbf{T}_a \mathbf{a}$$

$$\Sigma_p \leftarrow \mathbf{T}_s \Sigma_b \mathbf{T}_s^\top + \Sigma_s$$

$$\mathbf{K} \leftarrow \Sigma_p \mathbf{O}_s^\top \left(\mathbf{O}_s \Sigma_p \mathbf{O}_s^\top + \Sigma_o \right)^{-1}$$

$$\mu_b \leftarrow \mu_p + \mathbf{K}(\mathbf{o} - \mathbf{O}_s \mu_p)$$

$$\Sigma_b \leftarrow (\mathbf{I} - \mathbf{K} \mathbf{O}_s) \Sigma_p$$

$$\pi_{\text{LQG}}^*(b) = -K_{\text{LQR}} \mu_b$$

QMDP

POMDP Objective

$$\pi^* = \operatorname{argmax}_{\pi: B \rightarrow A} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t)) \right]$$

$$b' = \tau(b, a, o)$$

QMDP

POMDP Objective

$$\pi^* = \operatorname{argmax}_{\pi: B \rightarrow A} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t)) \right]$$

$$b' = \tau(b, a, o)$$

$$b' = \tau(b, a, o)$$

QMDP

POMDP Objective

$$\pi^* = \operatorname{argmax}_{\pi: B \rightarrow A} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t)) \right]$$

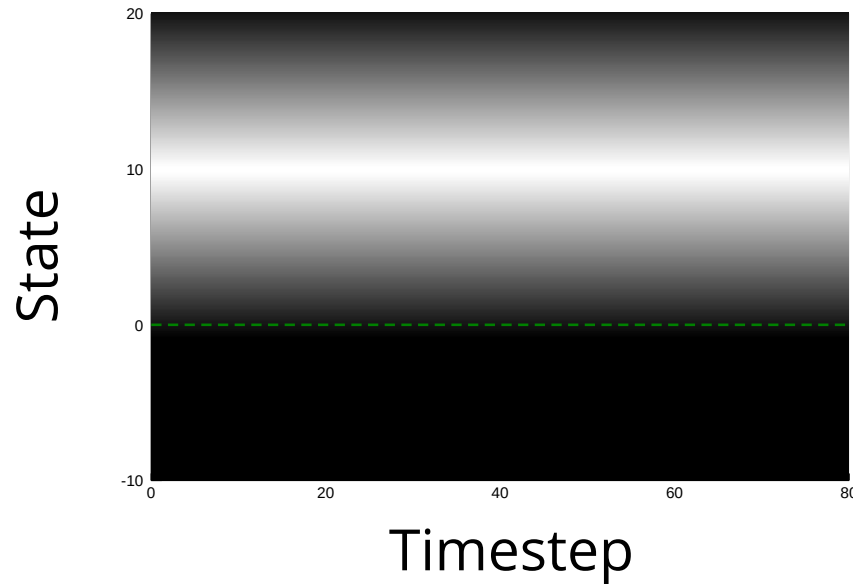
$$b' = \tau(b, a, o)$$

$$\pi_{\text{QMDP}}(b) = \operatorname{argmax}_{a \in A} \mathbb{E}_{s \sim b} [Q_{\text{MDP}}(s, a)]$$

$$b' = \tau(b, a, o)$$

Example: Tiger POMDP with Waiting

POMDP Example: Light-Dark



$$\mathcal{S} = \mathbb{Z}$$

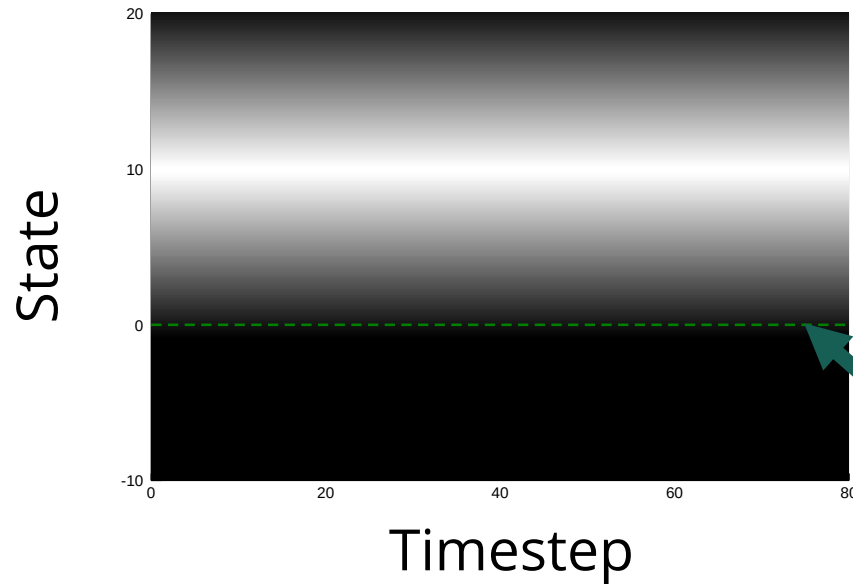
$$\mathcal{O} = \mathbb{R}$$

$$s' = s + a \quad o \sim \mathcal{N}(s, s - 10)$$

$$\mathcal{A} = \{-10, -1, 0, 1, 10\}$$

$$R(s, a) = \begin{cases} 100 & \text{if } a = 0, s = 0 \\ -100 & \text{if } a = 0, s \neq 0 \\ -1 & \text{otherwise} \end{cases}$$

POMDP Example: Light-Dark



$$\mathcal{S} = \mathbb{Z}$$

$$\mathcal{O} = \mathbb{R}$$

$$s' = s + a \quad o \sim \mathcal{N}(s, s - 10)$$

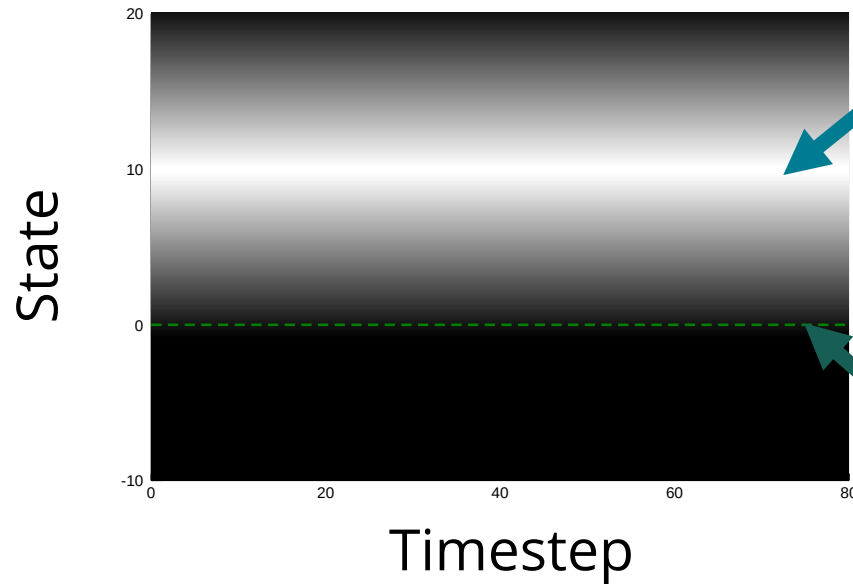
$$\mathcal{A} = \{-10, -1, 0, 1, 10\}$$

$$R(s, a) = \begin{cases} 100 & \text{if } a = 0, s = 0 \\ -100 & \text{if } a = 0, s \neq 0 \\ -1 & \text{otherwise} \end{cases}$$

Goal: $a = 0$ at $s = 0$

POMDP Example: Light-Dark

Accurate Observations



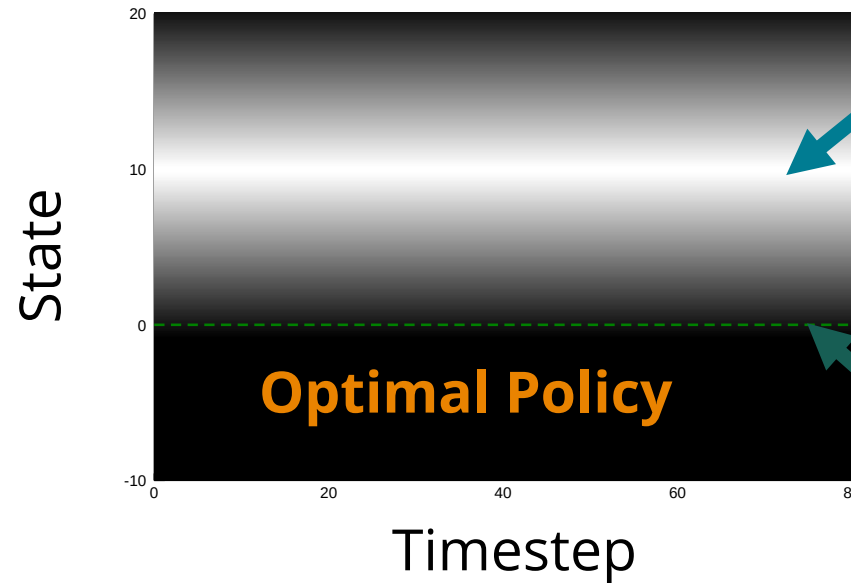
$$\begin{aligned}\mathcal{S} &= \mathbb{Z} & \mathcal{O} &= \mathbb{R} \\ s' &= s + a & o &\sim \mathcal{N}(s, s - 10) \\ \mathcal{A} &= \{-10, -1, 0, 1, 10\}\end{aligned}$$

$$R(s, a) = \begin{cases} 100 & \text{if } a = 0, s = 0 \\ -100 & \text{if } a = 0, s \neq 0 \\ -1 & \text{otherwise} \end{cases}$$

Goal: $a = 0$ at $s = 0$

POMDP Example: Light-Dark

Accurate Observations



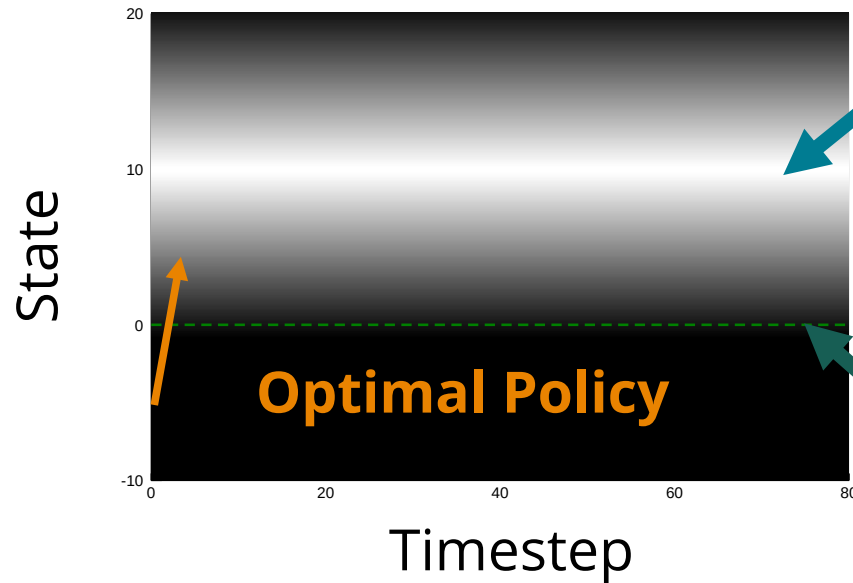
$$\begin{aligned}\mathcal{S} &= \mathbb{Z} & \mathcal{O} &= \mathbb{R} \\ s' &= s + a & o &\sim \mathcal{N}(s, s - 10) \\ \mathcal{A} &= \{-10, -1, 0, 1, 10\}\end{aligned}$$

$$R(s, a) = \begin{cases} 100 & \text{if } a = 0, s = 0 \\ -100 & \text{if } a = 0, s \neq 0 \\ -1 & \text{otherwise} \end{cases}$$

Goal: $a = 0$ at $s = 0$

POMDP Example: Light-Dark

Accurate Observations



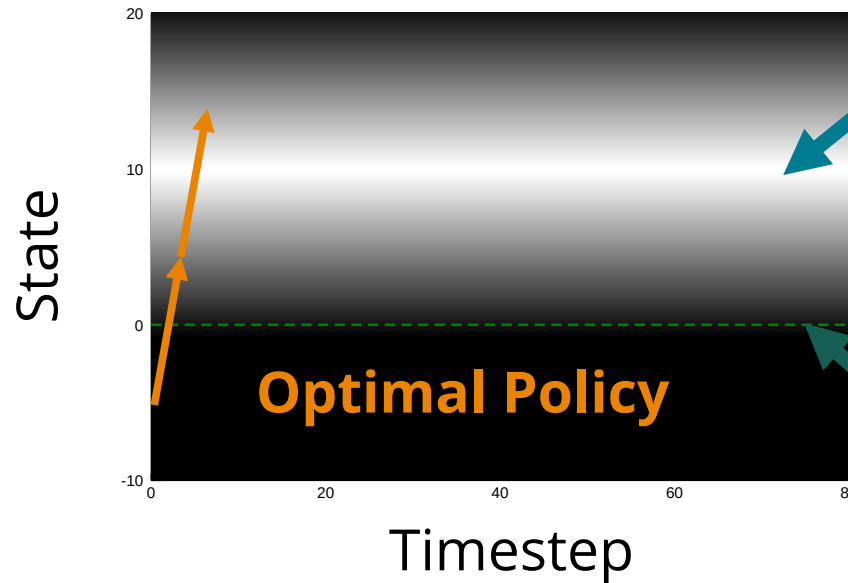
$$\begin{aligned}\mathcal{S} &= \mathbb{Z} & \mathcal{O} &= \mathbb{R} \\ s' &= s + a & o &\sim \mathcal{N}(s, s - 10) \\ \mathcal{A} &= \{-10, -1, 0, 1, 10\}\end{aligned}$$

$$R(s, a) = \begin{cases} 100 & \text{if } a = 0, s = 0 \\ -100 & \text{if } a = 0, s \neq 0 \\ -1 & \text{otherwise} \end{cases}$$

Goal: $a = 0$ at $s = 0$

POMDP Example: Light-Dark

Accurate Observations



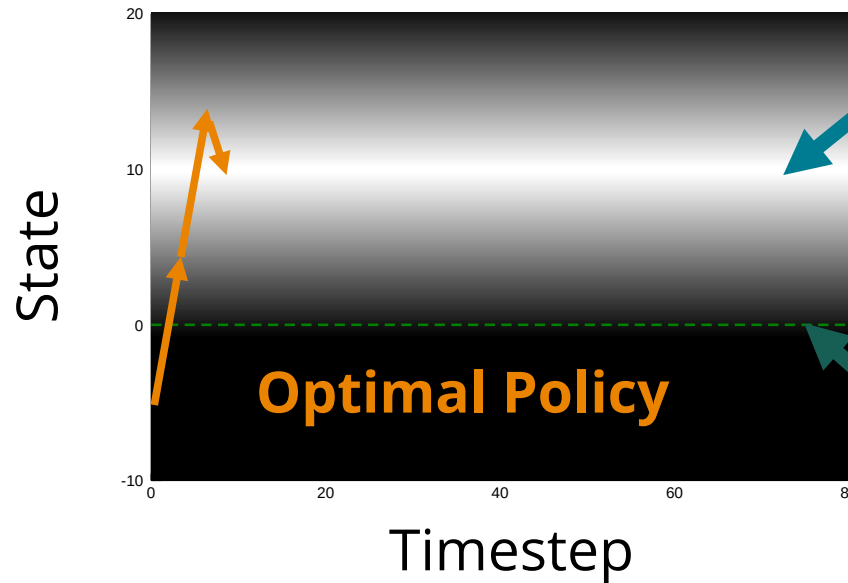
$$\begin{aligned}\mathcal{S} &= \mathbb{Z} & \mathcal{O} &= \mathbb{R} \\ s' &= s + a & o &\sim \mathcal{N}(s, s - 10) \\ \mathcal{A} &= \{-10, -1, 0, 1, 10\}\end{aligned}$$

$$R(s, a) = \begin{cases} 100 & \text{if } a = 0, s = 0 \\ -100 & \text{if } a = 0, s \neq 0 \\ -1 & \text{otherwise} \end{cases}$$

Goal: $a = 0$ at $s = 0$

POMDP Example: Light-Dark

Accurate Observations



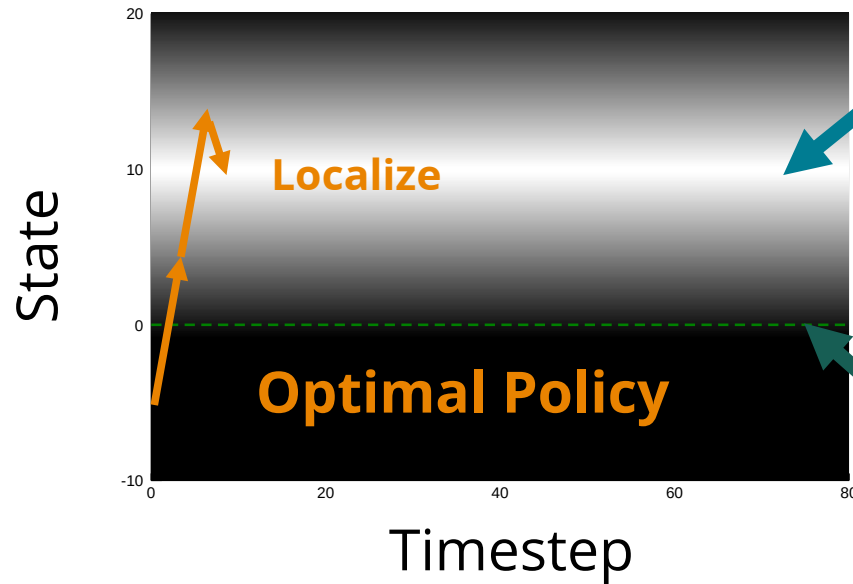
$$\begin{aligned}\mathcal{S} &= \mathbb{Z} & \mathcal{O} &= \mathbb{R} \\ s' &= s + a & o &\sim \mathcal{N}(s, s - 10) \\ \mathcal{A} &= \{-10, -1, 0, 1, 10\}\end{aligned}$$

$$R(s, a) = \begin{cases} 100 & \text{if } a = 0, s = 0 \\ -100 & \text{if } a = 0, s \neq 0 \\ -1 & \text{otherwise} \end{cases}$$

Goal: $a = 0$ at $s = 0$

POMDP Example: Light-Dark

Accurate Observations



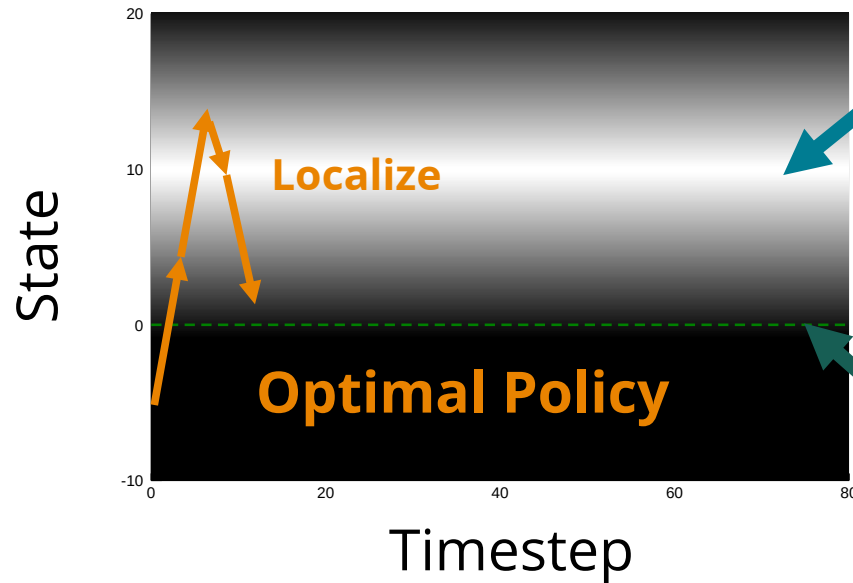
$$\begin{aligned}\mathcal{S} &= \mathbb{Z} & \mathcal{O} &= \mathbb{R} \\ s' &= s + a & o &\sim \mathcal{N}(s, s - 10) \\ \mathcal{A} &= \{-10, -1, 0, 1, 10\}\end{aligned}$$

$$R(s, a) = \begin{cases} 100 & \text{if } a = 0, s = 0 \\ -100 & \text{if } a = 0, s \neq 0 \\ -1 & \text{otherwise} \end{cases}$$

Goal: $a = 0$ at $s = 0$

POMDP Example: Light-Dark

Accurate Observations



$$\mathcal{S} = \mathbb{Z}$$

$$\mathcal{O} = \mathbb{R}$$

$$s' = s + a \quad o \sim \mathcal{N}(s, s - 10)$$

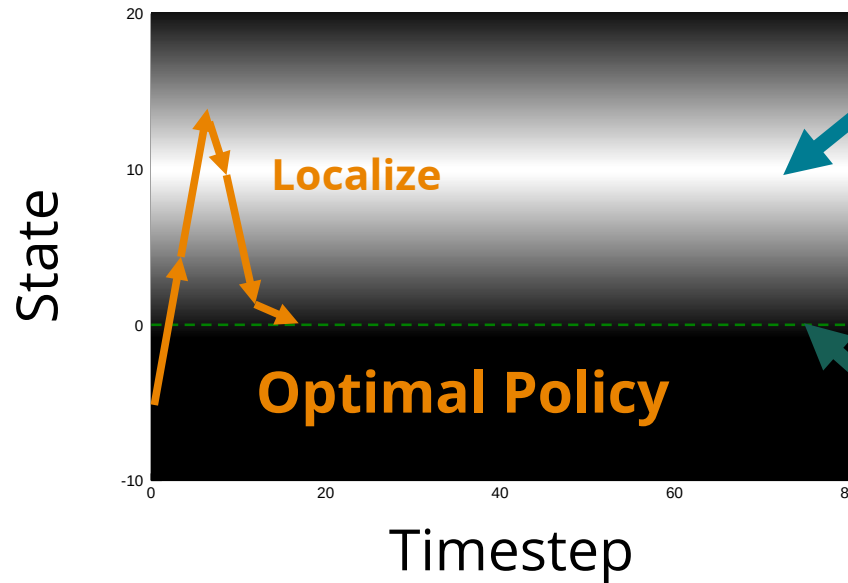
$$\mathcal{A} = \{-10, -1, 0, 1, 10\}$$

$$R(s, a) = \begin{cases} 100 & \text{if } a = 0, s = 0 \\ -100 & \text{if } a = 0, s \neq 0 \\ -1 & \text{otherwise} \end{cases}$$

Goal: $a = 0$ at $s = 0$

POMDP Example: Light-Dark

Accurate Observations



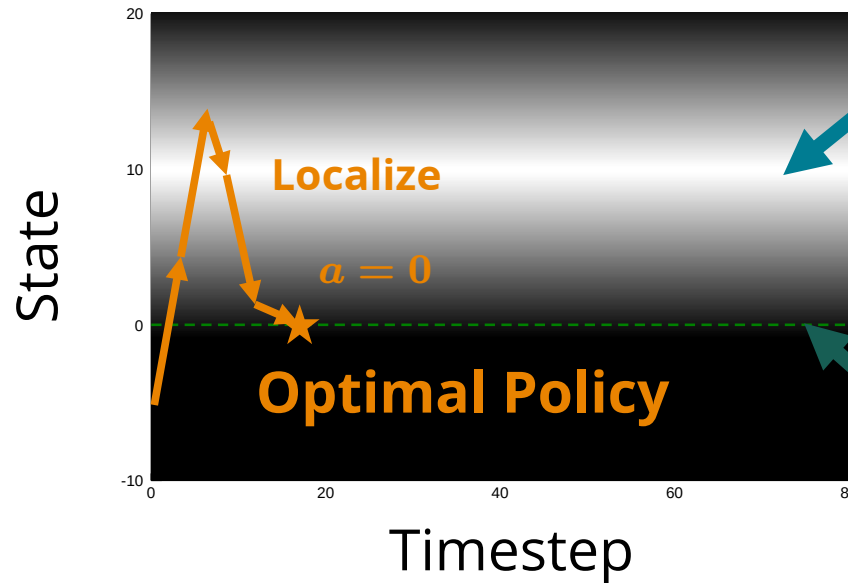
$$\begin{aligned}\mathcal{S} &= \mathbb{Z} & \mathcal{O} &= \mathbb{R} \\ s' &= s + a & o &\sim \mathcal{N}(s, s - 10) \\ \mathcal{A} &= \{-10, -1, 0, 1, 10\}\end{aligned}$$

$$R(s, a) = \begin{cases} 100 & \text{if } a = 0, s = 0 \\ -100 & \text{if } a = 0, s \neq 0 \\ -1 & \text{otherwise} \end{cases}$$

Goal: $a = 0$ at $s = 0$

POMDP Example: Light-Dark

Accurate Observations



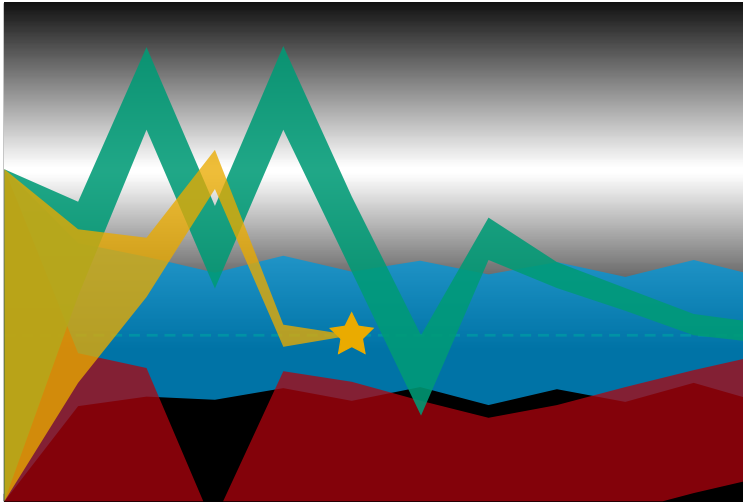
$$\begin{aligned}\mathcal{S} &= \mathbb{Z} & \mathcal{O} &= \mathbb{R} \\ s' &= s + a & o &\sim \mathcal{N}(s, s - 10) \\ \mathcal{A} &= \{-10, -1, 0, 1, 10\}\end{aligned}$$

$$R(s, a) = \begin{cases} 100 & \text{if } a = 0, s = 0 \\ -100 & \text{if } a = 0, s \neq 0 \\ -1 & \text{otherwise} \end{cases}$$

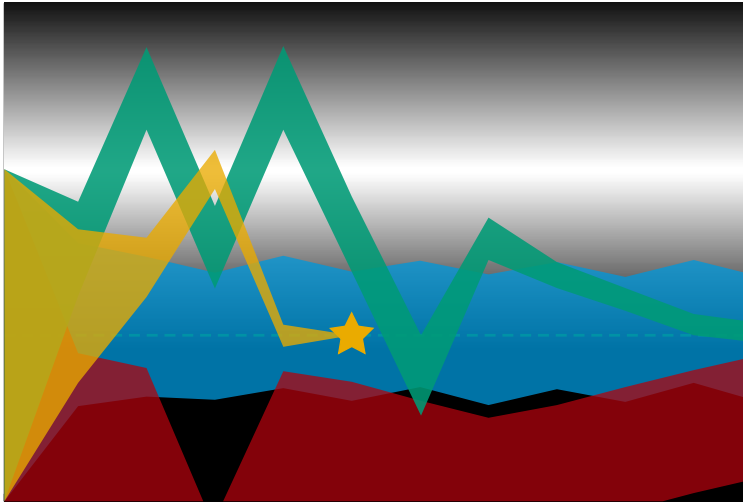
Goal: $a = 0$ at $s = 0$

POMDP

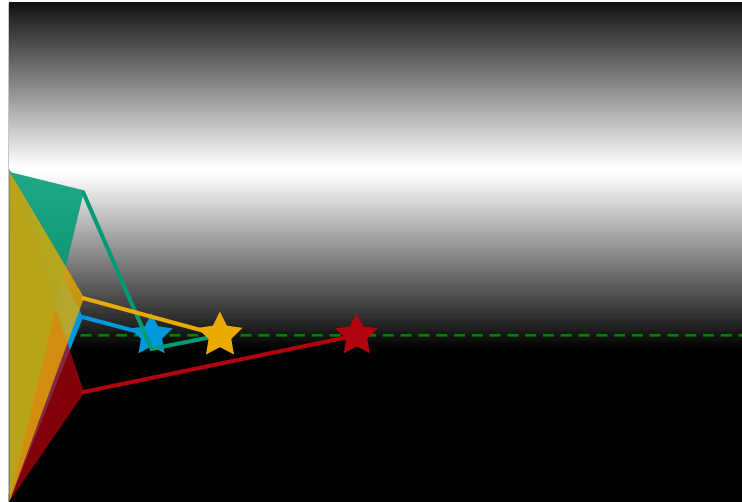
"Imagination"



POMDP "Imagination"

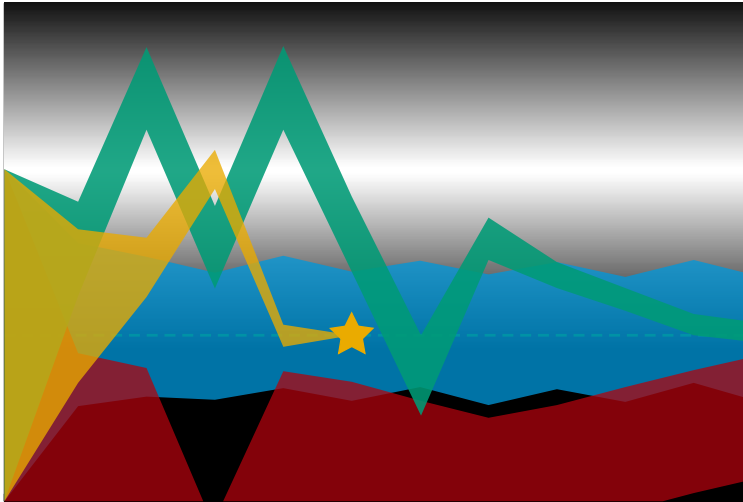


QMDP "Imagination"

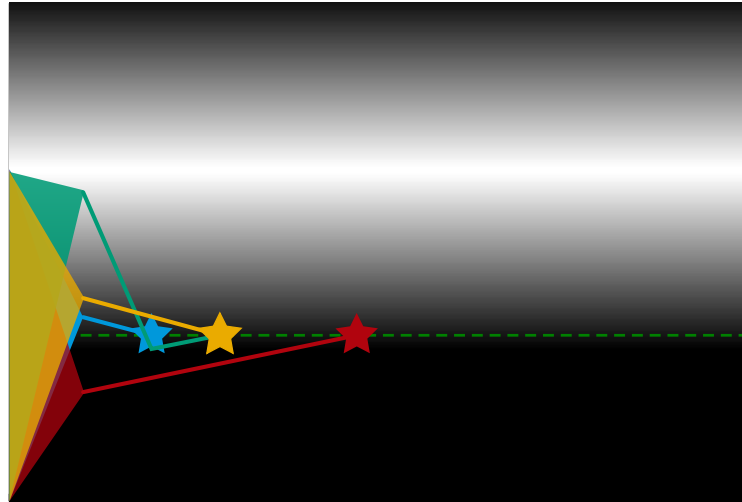


Same as **full observability**
on the next step

POMDP "Imagination"



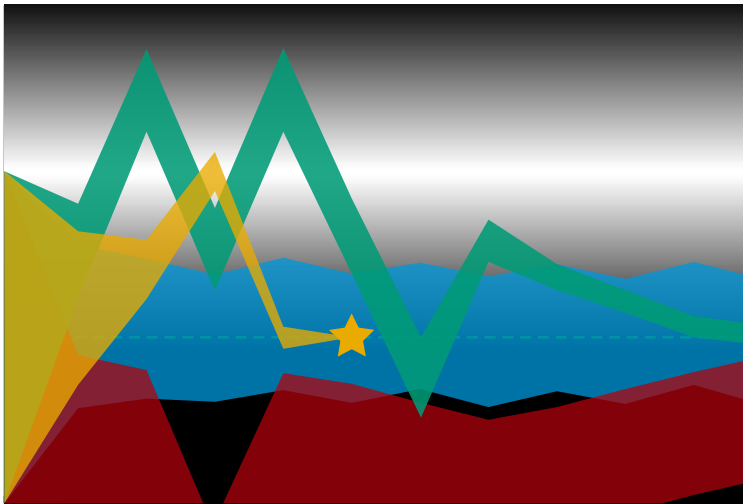
QMDP "Imagination"



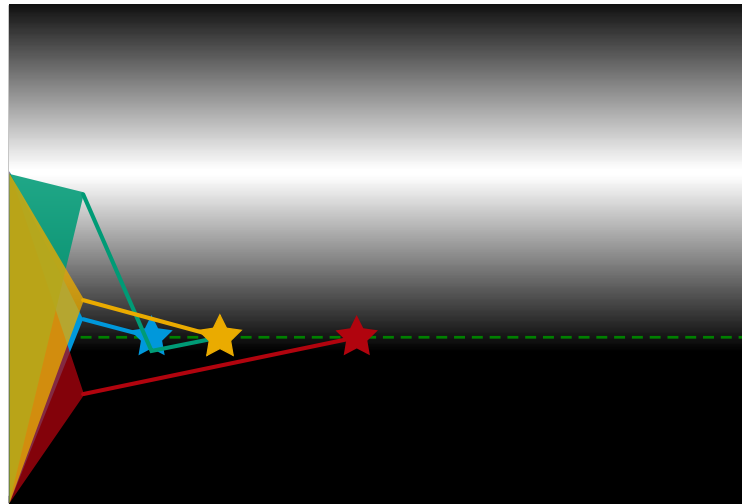
Same as **full observability**
on the next step

QMDP Behavior (Break)

POMDP "Imagination"

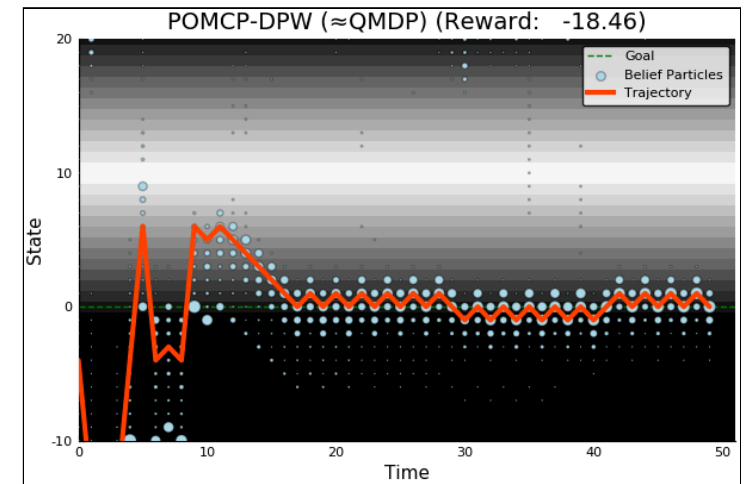


QMDP "Imagination"



Same as **full observability**
on the next step

QMDP Behavior (Break)



QMDP

Very effective for most POMDPs and cheap (same cost as solving MDP)

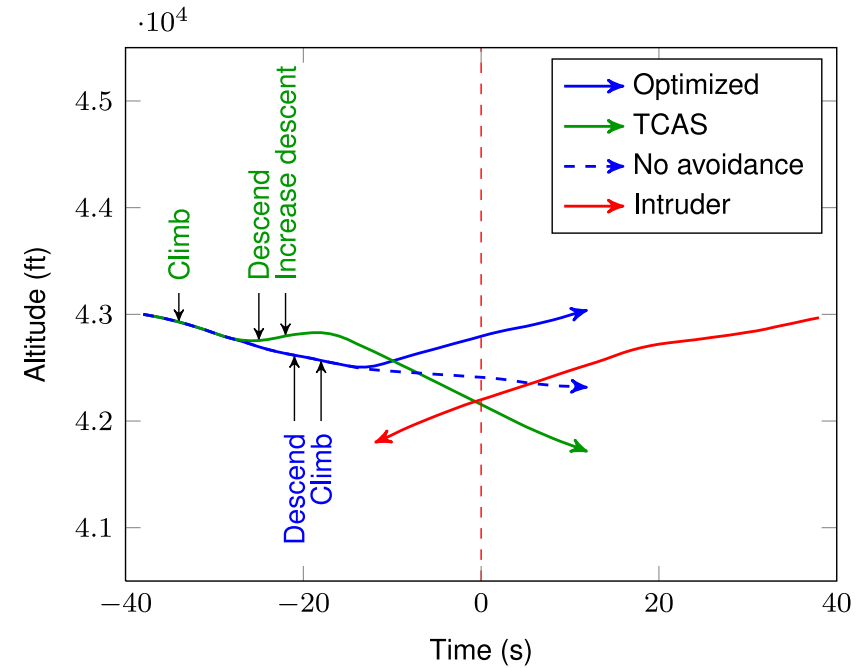
Two cases where it may not work well:

QMDP

Very effective for most POMDPs and cheap (same cost as solving MDP)

Used for ACAS X

[Kochenderfer, 2011]



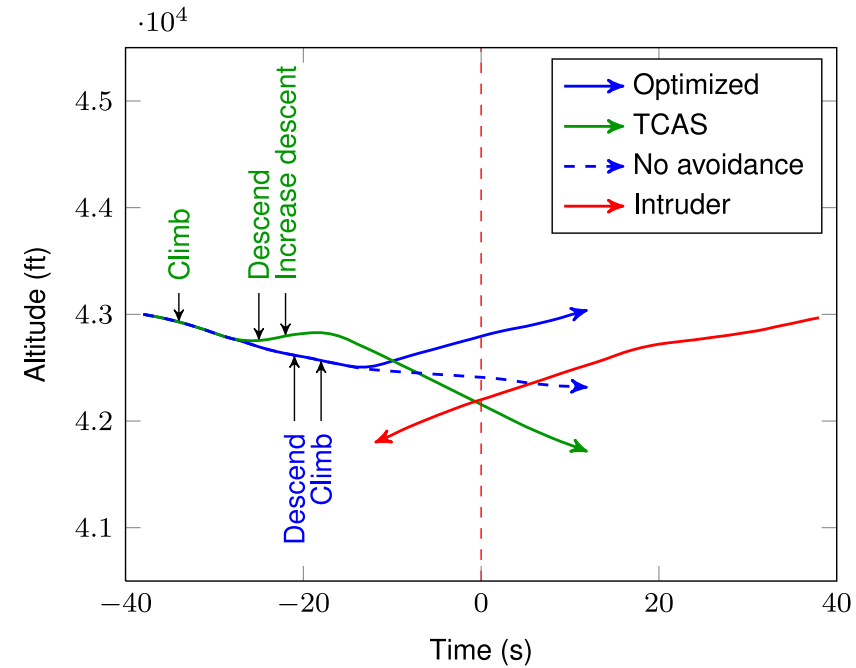
Two cases where it may not work well:

QMDP

Very effective for most POMDPs and cheap (same cost as solving MDP)

Used for ACAS X

[Kochenderfer, 2011]



Two cases where it may not work well:

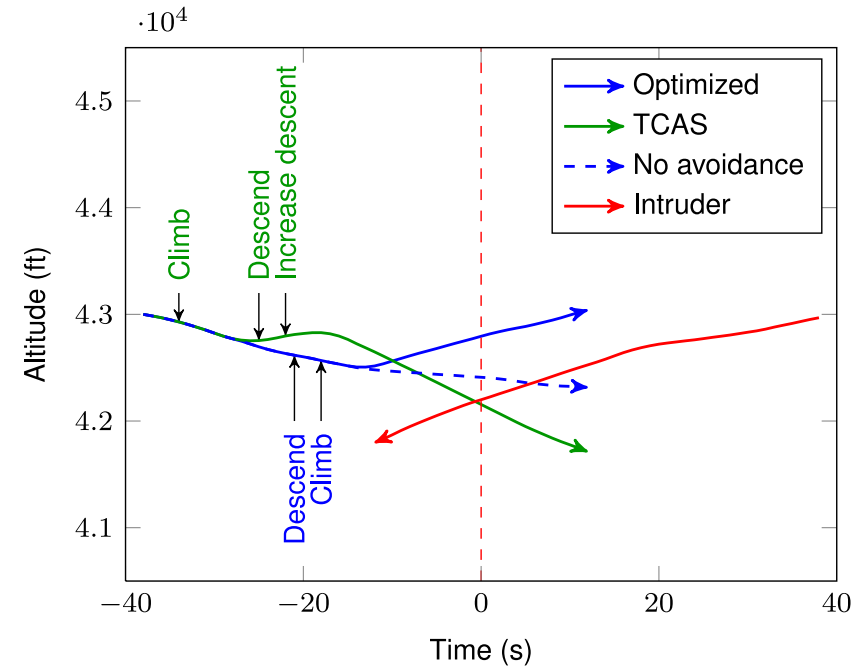
- *Costly* information gathering

QMDP

Very effective for most POMDPs and cheap (same cost as solving MDP)

Used for ACAS X

[Kochenderfer, 2011]



Two cases where it may not work well:

- *Costly* information gathering
- Long-lived uncertainty

Hindsight Optimization

POMDP Objective

$$\pi^* = \operatorname{argmax}_{\pi: B \rightarrow A} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t)) \right]$$

$$b' = \tau(b, a, o)$$

k-Markov

POMDP Objective

$$\pi^* = \operatorname{argmax}_{\pi: B \rightarrow A} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t)) \right]$$

$$b' = \tau(b, a, o)$$

Open Loop

A.K.A. Open Loop Feedback Control (OLFC)

POMDP Objective

$$\pi^* = \operatorname{argmax}_{\pi: B \rightarrow A} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t)) \right]$$

$$b' = \tau(b, a, o)$$